CSC4197

RESEARCH METHODOLOGY

INTRODUCTION TO RESEARCH & TYPES OF RESEARCH

2 WHAT IS RESEARCH?

- **Research** = systematic investigation to establish facts, develop new knowledge, or solve problems
- Derived from *recherche* (French): "to search again"
- Key properties: **systematic**, **objective**, **reproducible**, **evidence-based**
- Research ≠ searching Google, ≠ building a product
  - *Development*: applying known knowledge to build something
  - *Research*: generating **new** knowledge (even a negative result counts!)
- "If we knew what we were doing, it would not be called research." — A. Einstein

3 THE SCIENTIFIC METHOD (FOUNDATION OF ALL RESEARCH)

- **Observe** a phenomenon (e.g., "our app crashes more on low-RAM phones")
- **Question** it ("does memory pressure cause crash rate increase?")
- **Hypothesize** ("crash rate is proportional to memory pressure")
- **Experiment / collect data** (instrument the app, gather telemetry)
- **Analyze** (statistics, plots)
- **Conclude & communicate** (report, paper)
- **Iterate** — science is a loop, not a line

4 WHY SHOULD A CS STUDENT CARE ABOUT RESEARCH?

- Final-year **thesis/capstone** requires research skills
- Industry R&D roles (ML engineer, security researcher) demand them
- Graduate school admission = research statement + publications
- Even SWEs do "mini-research" daily: benchmarking, A/B testing, root-cause analysis
- **Example:** Google's PageRank began as a Stanford PhD research project — research became a trillion-dollar product

5 CLASSIFICATION AXIS 1: BASIC VS. APPLIED

	Basic (pure)	Applied
Goal	Expand fundamental knowledge	Solve a practical problem
Question	"Why/how does X happen?"	"How can we fix/improve Y?"
CS Example	Complexity of a new graph problem	Faster routing algorithm for Dhaka traffic app
Timeline	Long-term payoff	Short/medium-term payoff

Most undergraduate CS projects are **applied** — but they rest on basic research foundations.

6 CLASSIFICATION AXIS 2: QUALITATIVE VS. QUANTITATIVE VS. MIXED

- **Quantitative**: numeric data, statistical analysis, larger samples
  - *CS example:* measuring model accuracy across 5 datasets
- **Qualitative**: non-numeric (interviews, observations), meaning & context, smaller samples
  - *CS example:* interviewing 12 developers about why they ignore static-analysis warnings
- **Mixed-methods**: combines both — survey (quant) + follow-up interviews (qual)
  - *CS example:* usability study with task-completion times **and** think-aloud transcripts

7 CLASSIFICATION AXIS 3: PURPOSE

- **Exploratory** — "What is going on here?" (new, poorly understood areas)
- **Descriptive** — "What/how much/how many?" (characterize a phenomenon)
- **Explanatory (causal)** — "Why? Does X cause Y?" (test relationships)
- **Progression example (chatbot adoption in BD universities):**
  - Exploratory: interview students about how they use AI chatbots
  - Descriptive: survey 400 students — % usage, frequency, purposes
  - Explanatory: experiment — does chatbot tutoring *cause* higher quiz scores?

8 CS-SPECIFIC PARADIGM: DESIGN SCIENCE RESEARCH (DSR)

- Much of CS research **builds an artifact** (algorithm, tool, system, model) and **evaluates** it
- DSR cycle: identify problem → design artifact → build → **evaluate rigorously** → communicate
- Evaluation methods: benchmarking, controlled experiments, case studies, simulations
- **Example:** TrustICU-style work — build an ICU mortality prediction model (artifact), evaluate with AUROC/calibration on MIMIC-III (evaluation)

9 DECISION FRAMEWORK: WHEN TO USE WHICH TYPE?

- Ask these four questions:
- **How mature is the field?** New → exploratory/qualitative; mature → experimental
- **What does the RQ ask?** "What exists" → descriptive; "does X cause Y" → experiment
- **What data can you get?** No data access → simulation/secondary data; user access → survey/interview
- **Are you building something?** Yes → design science with empirical evaluation
- ⚠️ The research question chooses the method — never the reverse.

10 CASE STUDY: ONE TOPIC, FOUR RESEARCH TYPES

Topic: Mobile Financial Services (bKash-style) security

Research type	Study
Exploratory (qual)	Interview 15 users about perceived fraud risks
Descriptive (quant)	Survey 500 users: % who experienced phishing
Explanatory (experiment)	A/B test: does a warning UI reduce risky clicks?
Design science	Build + evaluate an SMS-phishing detection model

Same topic — completely different studies, skills, and papers.

11 HANDS-ON ACTIVITY (IN CLASS, ~30 MIN)

- **Task:** In pairs, classify these 4 real abstracts by:
- (a) basic/applied, (b) qual/quant/mixed, (c) exploratory/descriptive/explanatory/DSR
- **Sample items:**
  - "A Survey of Federated Learning Techniques" → ? https://www.mdpi.com/2073-431X/15/3/155
  - "Why Do Students Procrastinate on Programming Labs? A Grounded Theory Study" → ? https://onlinelibrary.wiley.com/doi/full/10.1002/pchj.70114
  - "BanglaBERT: A Pretrained Language Model for Bengali" → ? https://arxiv.org/abs/2608.05104
  - "Internet Performance Analysis of South Asian Countries Using End-to-End Internet Performance Measurements" → ? https://ieeexplore.ieee.org/abstract/document/8367431
- **Deliverable:** classification table + one-line justification each → submit in next class in hardcopy handwritten paper.

SCAN TO THE ABSTRACT









13 WRAP-UP, PROJECT KICKOFF & NEXT WEEK

- **Key takeaways**
- Research = systematic creation of new knowledge
- Classify by goal (basic/applied), data (qual/quant/mixed), purpose (explore/describe/explain), and DSR for artifacts
- The question drives the method
- 📢 **Semester project announced:** groups of 4–5; you will produce a 4–6 page IEEE-format mini-paper + presentation + mock viva by Week 12
- **Next week:** The research process & how to identify a research problem
- **Reading:** Keshav, "How to Read a Paper" (bring a printed/annotated copy) of the given paper in Teams or in your VUES portal.